Fermions & Spinors

To introduce spinors, let's back up & think more carefully about how Lorentz (& Poincaré) invariance is realized on field theories.

For a real scalar field,

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu} \partial_\mu \phi(x) \partial_\nu \phi(y) - \frac{1}{2} m^2 \phi^2(x) + \dots$$

$$\rightarrow \left(g^{\mu\nu} \partial_\mu \partial_\nu - m^2 \right) \phi(x) = 0 + \dots$$

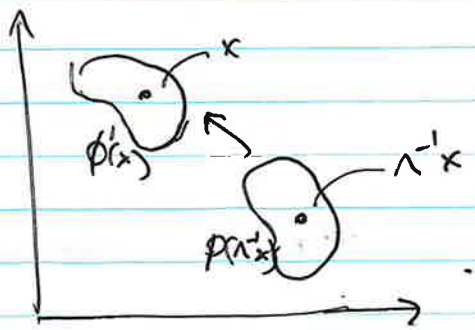
Consider performing an <u>active</u> Lorentz transformation. This means we leave the coordinate system (reference frame) fixed and physically rotate/boost the field by some angle/velocity.

The Lorentz transf. law for the scalar field is:

$$\phi(x) \longrightarrow \phi'(x) = \phi(\Lambda^{-1} x)$$

ϕ transf. field at x = original field at original point $(\Lambda^{-1}x)$

Visually:



Under this transformation,

$$\partial_\mu \phi(x) \to \partial_\mu \phi'(x) = \partial_\mu \phi(\Lambda^{-1}x) = (\Lambda^{-1})^\nu{}_\mu (\partial_\nu \phi)(\Lambda^{-1}x)$$

$$\to$$

$$g^{\mu\nu}\partial_\mu\phi(x)\partial_\nu\phi(x) \to g^{\mu\nu}\partial_\mu\phi'(x)\partial_\nu\phi'(x)$$

$$= g^{\mu\nu}\left[(\Lambda^{-1})^\rho{}_\mu(\partial_\rho\phi)(\Lambda^{-1}x)\right]\left[(\Lambda^{-1})^\sigma{}_\nu(\partial_\sigma\phi)(\Lambda^{-1}x)\right]$$

$$= \underbrace{\left((\Lambda^{-1})^\rho{}_\mu(\Lambda^{-1})^\sigma{}_\nu g^{\mu\nu}\right)}_{=g^{\rho\sigma}} \partial_\rho\phi(\Lambda^{-1}x)\partial_\sigma\phi(\Lambda^{-1}x)$$

$$= g^{\rho\sigma}\partial_\rho\phi(\Lambda^{-1}x)\partial_\sigma\phi(\Lambda^{-1}x)$$

The mass term is left similarly invariant, up to the fields being evaluated at $\Lambda^{-1}x$.

Thus:

$$\mathcal{L}(x) \mapsto \mathcal{L}(\Lambda^{-1}x)$$

So the ~~det~~ Lagrangian is a <u>scalar</u> and the action is Lorentz invariant.

$$S = \int d^4x\, \mathcal{L}(x) = \int d^4(\Lambda^{-1}x)\, \mathcal{L}(\Lambda^{-1}x)$$

The scalar field has the simplest possible transformation under Lorentz (Poincaré) transformations.

~~We will now see that we get more~~

As a result, it had excitations corresp. to <u>spin-0 particles</u>

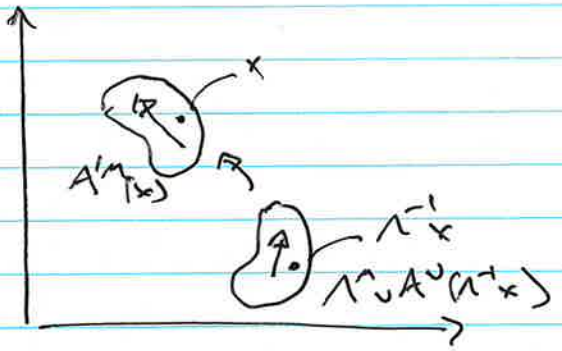
We'll now see that more interesting representations of the Poincaré group lead to particle excitations with non-trivial spin/helicity quantum numbers.

As a first example, consider a vector field: $A^\mu(x)$

This carries not only a value but an orientation in spacetime: under a Lorentz transf.: (active)

$$A^\mu(x) \longmapsto A'^\mu(x) = \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x)$$

→ in addition to the transformed point, we must also rotate/boost the orientation:



We'll return to a Lorentz-invariant Lagrangian for vector fields later, and see that their excitations lead to spin-1 bosons.

Unfortunately, this brings in the complications of gauge invariance.

But this is enough to try and generalize the arguments.

Consider a field Φ_a w/ n components ($a = 1, 2, \ldots n$)
(i.e. for the scalar, $n=0$ while for the vector $n=4$)

Under a Lorentz transformation:

$$\Phi_a(x) \xrightarrow{\Lambda} \Phi'_a(x) = \left[M_R(\Lambda)\right]_{ab} \Phi_b(\Lambda^{-1}x)$$

We'll frequently suppress the indices & arguments of the fields.

$$\Phi \to M_R(\Lambda)\Phi$$

The matrices $M_R(\Lambda)$ furnish an $n \times n$-dimensional representation of the Lorentz group. They satisfy all the properties of the group structure w/ matrix multiplication (closure, inverse, identity, associativity). E.g., under successive transformations, Λ and Λ', we get a new matrix.

$$\Phi \to M_R(\Lambda')M_R(\Lambda)\Phi = M_R(\Lambda'')\Phi$$

So our goal is to find the n-dimensional representations.

We already have $n=0$ and $n=4$.

$n=0$: $M_R(\Lambda) = 1$.

$n=4$: $\left[M_R(\Lambda)\right]^\mu{}_\nu = \Lambda^\mu{}_\nu$

To get other representations, recall that representations of finite Lorentz transformations are built by exponentiating the generators: i.e., we build them from successive infinitesimal transformations:

$$\Phi_a \longrightarrow \left[\delta_{ab} - \frac{i}{2}\omega_{\mu\nu}(J_R^{\mu\nu})_{ab} \right] \Phi_b$$

$\omega_{\mu\nu}$: infinitesimal param; $J_R^{\mu\nu}$: generators in rep. R associated w/ Φ

We already know the generators satisfy:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i\left(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho}\right)$$

The finite transf. in the representation R is given by:

$$\Phi_a \longrightarrow M_R(\Lambda)_{ab}\Phi_b = \exp\left[\frac{-i}{2}\omega_{\mu\nu}J_R^{\mu\nu}\right]_{ab}\Phi_b$$

We already saw this for the vector field in the homework:

$$(J^{\mu\nu})^{\alpha}_{\ \beta} = i\left(g^{\mu\alpha}\delta^{\nu}_{\ \beta} - g^{\nu\alpha}\delta^{\mu}_{\ \beta}\right)$$

and a corresponding finite transf. obtained by (matrix) exponentiation.

Now we're going to find another, more surprising representation.

The Dirac Equation

Our next – extremely important – example is the spin-$1/2$ Dirac fermion field. Our first goal is to find the generators of Lorentz transformations for this representation. This will tell us the dimensionality of the field and help fix some of its properties. We'll then write down the appropriate relativistic field equation — the Dirac equation — and find an action for the field we can quantize. This will yield fermions we can use in constructing our theory.

With the benefit of a great deal of hindsight, we can introduce the new representation as follows.

First, introduce a set of four $n \times n$ matrices denoted γ^μ ($\mu = 0, 1, 2, 3$) called the Dirac γ-matrices, which satisfy the following anti-commutation relations:

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2 g^{\mu\nu} \cdot \mathbb{1}_{n\times n} \qquad (*)$$

The curly brackets distinguish anti-commutation from commutation. Note that for $\mu \neq \nu$, $g^{\mu\nu} = 0$ so swapping the order of two γ-matrices introduces a minus sign:

$$\gamma^\mu \gamma^\nu = -\gamma^\nu \gamma^\mu \quad (\mu \neq \nu)$$

The relation (*) is known as the "Dirac algebra" or "Clifford algebra".

Given a set of γ matrices that satisfy (*), one can show that an n-dimensional representation of the Lorentz algebra is given by:

$$J_D^{\mu\nu} \equiv S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$$

(sometimes $S^{\mu\nu}$ is denoted $\Sigma^{\mu\nu}$ or $\sigma^{\mu\nu}$)

I won't do the algebra here, but the point is just that — with repeated use of the anticommutation relation — one can show $S^{\mu\nu}$ satisfies

$$[S^{\mu\nu}, S^{\rho\sigma}] = i(g^{\nu\rho}S^{\mu\sigma} - g^{\mu\rho}S^{\nu\sigma} - g^{\nu\sigma}S^{\mu\rho} + g^{\mu\sigma}S^{\nu\rho})$$

— i.e., the Lorentz algebra. This establishes that it indeed is a representation.

So what four linearly independent matrices $\gamma^0, \gamma^1, \gamma^2, \gamma^3$ do the job? The simplest representation turns out to be four-dimensional, w/ one set being:

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

These are written in 2×2 block-form, w/ each block a 2×2 matrix, explicitly:

$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$\sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

(i.e., the Pauli matrices)

It is sometimes useful to use the shorthand $\sigma^\mu = (\mathbb{1}_{2\times 2}, \vec{\sigma})$ and $\bar{\sigma}^\mu = (\mathbb{1}_{2\times 2}, -\vec{\sigma})$ so that

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}$$

This set is known as the "Weyl" or "Chiral" representation.

There are other choices for the 4×4 matrices, which are just a different choice of basis. They're related to the above via a similarity transformation. Physics ~~shouldn't~~ should never depend on the choice of representation.

One can work out explicit forms of the rotation & boost generators in this representation by using the properties of the Pauli matrices, particularly,

$$[\sigma^i, \sigma^j] = 2i\epsilon^{ijk}\sigma^k$$

Since the generators are 4×4 matrices, the fields they act on must have four components.

We are thus led to introduce a spinor field $\psi(x)$ which has index structure,

$$\psi_a(x) \quad , \quad (\gamma^\mu)_{ab} \qquad a,b = 1,2,3,4$$

So that the Lorentz transformation law for $\psi_a(x)$ is:

$$\psi_a(x) \longrightarrow \psi'_a(x) = \left[M_D(\Lambda)\right]_{ab} \psi_b(\Lambda^{-1}x)$$

$$= \exp\left[\frac{-i}{2}\omega_{\mu\nu} S^{\mu\nu}\right]_{ab} \psi_b(\Lambda^{-1}x)$$

$$\simeq \left[\delta_{ab} - \frac{i}{2}\omega_{\mu\nu}(S^{\mu\nu})_{ab}\right]\psi_b(\Lambda^{-1}x)$$

I will almost always suppress the spinor indices in what follows.

Now, we would like to find a field equation for $\psi(x)$ with the right Lorentz transformation properties.

With a little work, one can show that

$$\left[\gamma^\mu, S^{\rho\sigma}\right] = (J^{\rho\sigma})^\mu{}_\nu \gamma^\nu$$

w/ $(J^{\rho\sigma})^\mu{}_\nu$ the generator of the vector representation,

of the Lorentz group,

$$(\mathcal{J}^{\rho\sigma})^{\mu}{}_{\nu} = i\left(g^{\rho\mu}\delta^{\sigma}{}_{\nu} + g^{\sigma\mu}\delta^{\rho}{}_{\nu}\right)$$

To first order in infinitesimal $\omega_{\rho\sigma}$, the commutator can be equivalently rewritten as:

$$\left(1 + \tfrac{i}{2}\omega_{\rho\sigma}S^{\rho\sigma}\right)\gamma^{\mu}\left(1 - \tfrac{i}{2}\omega_{\rho\sigma}S^{\rho\sigma}\right)$$

$$= \left(1 - \tfrac{i}{2}\omega_{\rho\sigma}\mathcal{J}^{\rho\sigma}\right)^{\mu}{}_{\nu}\gamma^{\nu}$$

This is just the infinitesimal form of

$$\left\{\exp\left[\tfrac{-i}{2}\omega_{\rho\sigma}S^{\rho\sigma}\right]\right\}^{-1}\gamma^{\mu}\left\{\exp\left[\tfrac{-i}{2}\omega_{\rho\sigma}S^{\rho\sigma}\right]\right\}$$

$$= \exp\left[\tfrac{-i}{2}\omega_{\rho\sigma}\mathcal{J}^{\rho\sigma}\right]^{\mu}{}_{\nu}\gamma^{\nu}$$

or simply,

$$M_D^{-1}(\Lambda)\,\gamma^{\mu}\,M_D(\Lambda) = \Lambda^{\mu}{}_{\nu}\gamma^{\nu}$$

This is the statement that the γ-matrices themselves transform as a vector under Lorentz transformations.

Therefore, we can contract γ^{μ} w/ a derivative ∂_{μ} to form a Lorentz-invariant differential operator.

With this knowledge, we can boldly propose the following field equation for $\psi(x)$:

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0$$

This is the Dirac equation.

We can check how this equation behaves under a Lorentz transformation:

$$[i\gamma^\mu \partial_\mu - m]\psi(x) \rightarrow [i\gamma^\mu \partial'_\mu - m]\psi'(x)$$

$$= [i\gamma^\mu \partial_\mu - m] M_D \psi(\Lambda^{-1}x)$$

$$= \{[i\gamma^\mu (\Lambda^{-1})^\nu{}_\mu \partial_\nu - m] M_D \psi\}(\Lambda^{-1}x)$$

$$= \{M_D M_D^{-1} [i\gamma^\mu (\Lambda^{-1})^\nu{}_\mu \partial_\nu - m] M_D \psi\}(\Lambda^{-1}x)$$

$$= M_D \{i M_D^{-1} \gamma^\mu M_D (\Lambda^{-1})^\nu{}_\mu \partial_\nu - m M_D^{-1} M_D\} \psi(\Lambda^{-1}x)$$

$$= M_D \{i\gamma^\rho (\Lambda^{-1})^\nu{}_\mu \Lambda^\mu{}_\rho \partial_\nu - m\} \psi(\Lambda^{-1}x)$$

$$= M_D [i\gamma^\nu \partial_\nu - m] \psi(\Lambda^{-1}x) = 0$$

That is, the transformed field satisfies the same equation.

One reason this is a good choice for an equation of motion is that the Dirac equation implies the Klein-Gordon equation for $\psi(x)$:

Starting w/ the Dirac equation, multiply both sides by $(-i\gamma^\mu \partial_\mu - m)$:

$$0 = (-i\gamma^\mu \partial_\mu - m)(i\gamma^\nu \partial_\nu - m)\psi(x)$$

$$= (\gamma^\mu \gamma^\nu \partial_\mu \partial_\nu + m^2)\psi(x)$$

$$= \left(\frac{1}{2}\{\gamma^\mu, \gamma^\nu\}\partial_\mu \partial_\nu + m^2\right)\psi(x)$$

$$= (g^{\mu\nu}\partial_\mu \partial_\nu + m^2)\psi(x)$$

(split first term in half, relabel the indices, then commute partial derivatives)

Thus the classical field will admit wave-like solutions we are interested in. The first order equation is more powerful, as it interweaves the spin-dependence which is non-trivial.

Now we need to construct an action from which the Dirac equation arises as an equation of motion.

To do so, we need to form a Lorentz-invariant quantity involving two spinors (so that we get an equation for ψ after differentiation)

The obvious candidate is $\psi^\dagger\psi$. Under a Lorentz transformation:

$$\psi(x) \mapsto M_D(\Lambda)\psi(\Lambda^{-1}x)$$

$$\psi^\dagger(x) \mapsto \psi^\dagger(\Lambda^{-1}x) M_D^\dagger(\Lambda)$$

so

$$\psi^\dagger\psi \mapsto \psi^\dagger M_D^\dagger M_D \psi .$$

If $M_D(\Lambda)$ was unitary, this would work ($\Rightarrow M_D^\dagger = M_D^{-1}$)

However, the generator of boosts S^{0i} is not Hermitian, so Lorentz transformations are not generally unitary, so $\psi^\dagger\psi$ is not Lorentz invariant.

This can be remedied by finding a relationship b/w $M_D^\dagger$ and M_D^{-1}. First, note from the explicit form of the γ-matrices that*

$$(\gamma^0)^\dagger = \gamma^0 \quad , \quad (\gamma^i)^\dagger = -\gamma^i$$

This is equivalent to: $\gamma^0\gamma^\mu\gamma^0 = (\gamma^\mu)^\dagger$

* you should be suspicious of arguments that rely on a choice of representation! In general, one could introduce a new matrix A which conjugates the γ-matrices and is only <u>numerically</u> equal to γ^0. See Appendix G of Dreiner, Haber, Martin hep-ph [0812.1594] for details.

We can check this explicitly:

$$\mu = 0: \quad \gamma^0 \gamma^0 \gamma^0 = \frac{1}{2}\{\gamma^0, \gamma^0\}\gamma^0 = \frac{1}{2}(2g^{00})\gamma^0 = \gamma^0 = (\gamma^0)^\dagger$$

$$\mu = i: \quad \gamma^0 \gamma^i \gamma^0 = -\gamma^0 \gamma^0 \gamma^i = -\gamma^i = (\gamma^i)^\dagger$$

Therefore, the adjoint of the generator is:

$$(S^{\mu\nu})^\dagger = \left(\frac{i}{4}[\gamma^\mu, \gamma^\nu]\right)^\dagger = \frac{-i}{4}[(\gamma^\mu)^\dagger, (\gamma^\nu)^\dagger]$$

$$= \frac{-i}{4}[\gamma^0\gamma^\nu\gamma^0, \gamma^0\gamma^\mu\gamma^0]$$

$$= \gamma^0\left(\frac{i}{4}[\gamma^\mu, \gamma^\nu]\right)\gamma^0 = \gamma^0 S^{\mu\nu}\gamma^0$$

The adjoint of the Lorentz-transformation matrix is thus:

$$M_D^\dagger(\Lambda) = \exp\left[\frac{i}{2}\omega_{\mu\nu}(S^{\mu\nu})^\dagger\right] = \gamma^0 \exp\left[\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}\right]\gamma^0$$

$$= \gamma^0 M_D^{-1}(\Lambda)\gamma^0$$

Armed with this, we ~~construct~~ define the <u>Dirac conjugate</u> spinor:

$$\overline{\psi}(x) \equiv \psi^\dagger(x)\gamma^0$$

Under a Lorentz transformation,

$$\bar{\psi}(x) \mapsto \psi^\dagger M_D^\dagger \gamma^0 = \psi^\dagger (\gamma^0 M_D^{-1} \gamma^0)\gamma^0$$
$$= \bar{\psi} M_D^{-1}$$

Thus the product $\bar{\psi}\psi$ is <u>Lorentz invariant</u>:

$$\bar{\psi}\psi \mapsto \bar{\psi} M_D^{-1} M_D \psi = \bar{\psi}\psi$$

And is a Lorentz scalar.
We can form other bi-linears: for example,

$$\bar{\psi}\gamma^\mu\psi \mapsto \bar{\psi}(M_D^{-1}\gamma^\mu M_D)\psi = \Lambda^\mu{}_\nu \bar{\psi}\gamma^\nu\psi$$
$$= \Lambda^\mu{}_\nu \gamma^\nu$$

So the quantity $\bar{\psi}\gamma^\mu\psi$ transforms as a 4-vector.

From these we can construct the <u>Dirac Lagrangian</u>:

$$\boxed{\mathcal{L} = \bar{\psi} i\gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi}$$

To treat this Lagrangian, we should regard ψ and $\bar{\psi}$ as distinct field variables (we would do the same for a complex scalar, φ and $\varphi^\dagger$).

Then, applying the principle of least action, we can vary $\mathcal{L}$ with respect to both ψ and $\bar{\psi}$:

$$0 = \delta S = \int d^4x \left\{ \delta\bar{\psi} \left[i\gamma^\mu \partial_\mu - m \right] \psi + \bar{\psi} \underbrace{\left[i\gamma^\mu \partial_\mu - m \right]}_{\text{integrate by parts}} \delta\psi \right\}$$

$$= \int d^4x \, \partial_\mu (\bar{\psi} i\gamma^\mu \psi) + \int d^4x \, \delta\bar{\psi} \left[i\gamma^\mu \partial_\mu - m \right] \psi$$

$$+ \cancel{\int d^4x \, \bar{\psi} \left[-i\gamma^\mu \partial_\mu - m \right]}$$

$$+ \int d^4x \left[-i \partial_\mu \bar{\psi} \gamma^\mu - m\bar{\psi} \right] \delta\psi$$

This gives us two equations:

$$\left[i\gamma^\mu \partial_\mu - m \right] \psi = 0$$

$$\left[-i \partial_\mu \bar{\psi} \gamma^\mu - m\bar{\psi} \right] = 0$$

These are just the Dirac* equation and its Dirac conjugate.

We've arrived at a Lagrangian that yields our desired classical equation of motion. This is a good starting point for quantizing the theory.

Weyl Spinors

Before we find classical solutions to quantize, let's unpack the Dirac theory a bit more.

While the Dirac representation was discovered first (in physics, at least) and is extremely important in its own right, it is not the simplest non-trivial spinor representation of the Lorentz group. In fact, the Dirac representation is reducible: note that the generators of rotations & boosts are block diagonal:

$$S^{0i} = \frac{-i}{2}\begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}, \quad S^{ij} = \frac{1}{2}\epsilon^{ijk}\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}$$

We can thus decompose the Dirac representation into two irreducible representations, each acting separately on two, 2-component Weyl spinors:

$$\psi(x) = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

ψ_L and ψ_R are each a 2-component field. We can work out how they transform under a Lorentz transformation.

Starting from:

$$\psi \to \exp\left[-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}\right]\psi$$

$$= \exp\left[-i\omega_{0i}S^{0i} - \frac{i}{2}\omega_{ij}S^{ij}\right]\psi$$

$$= \exp\left[-i\omega_{0i}\left(\frac{-i}{2}\right)\begin{pmatrix}\sigma^i & 0\\ 0 & -\sigma^i\end{pmatrix} - \frac{i}{2}\omega_{ij}\frac{1}{2}\epsilon^{ijk}\begin{pmatrix}\sigma^k & 0\\ 0 & \sigma^k\end{pmatrix}\right]\psi$$

We define boost and rotation parameters:

$$\beta^i = \omega_{0i}\,, \quad \theta^k = \frac{1}{2}\epsilon^{ijk}\omega_{ij}$$

and see:

$$\begin{pmatrix}\psi_L\\ \psi_R\end{pmatrix} \to \exp\begin{bmatrix}-\vec{\beta}\cdot\frac{\vec{\sigma}}{2} - i\vec{\theta}\cdot\frac{\vec{\sigma}}{2} & 0\\ 0 & +\vec{\beta}\cdot\frac{\vec{\sigma}}{2} - i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}\end{bmatrix}\begin{pmatrix}\psi_L\\ \psi_R\end{pmatrix}$$

So:

$$\psi_L \to \exp\left[-\vec{\beta}\cdot\frac{\vec{\sigma}}{2} - i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}\right]\psi_L$$

$$\psi_R \to \exp\left[+\vec{\beta}\cdot\frac{\vec{\sigma}}{2} - i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}\right]\psi_R$$

or, infinitesimally:

$$\psi_L \to \left(1 - \vec{\beta}\cdot\frac{\vec{\sigma}}{2} - i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}\right)\psi_L$$

$$\psi_R \to \left(1 + \vec{\beta}\cdot\frac{\vec{\sigma}}{2} - i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}\right)\psi_R$$

Thus, ψ_L and ψ_R transform the same under rotations, but oppositely under boosts.

We can expand the Dirac equation in terms of ψ_L and ψ_R:

$$(i\gamma^\mu \partial_\mu - m)\psi = \left\{ i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\partial_0 + i\begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}\partial_i - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} m \right\}\begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

$$= \begin{pmatrix} -m & i(\partial_0 + \vec{\sigma}\cdot\vec{\nabla}) \\ i(\partial_0 - \vec{\sigma}\cdot\vec{\nabla} & -m \end{pmatrix}\begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = 0$$

Or,

$$i(\partial_0 - \vec{\sigma}\cdot\vec{\nabla})\psi_L - m\psi_R = 0$$

$$i(\partial_0 + \vec{\sigma}\cdot\vec{\nabla})\psi_R - m\psi_L = 0$$

We see that the mass term mixes, or couples, the two Weyl fields ψ_L and ψ_R. In the limit $m \to 0$ (or, in the high-energy limit when masses can be ignored!) these reduce to the <u>Weyl</u> equations:

$$i(\partial_0 - \vec{\sigma}\cdot\vec{\nabla})\psi_L = 0 \quad , \quad i(\partial_0 + \vec{\sigma}\cdot\vec{\nabla})\psi_R = 0$$

Using the shorthand

$$\sigma^\wedge = (1, \vec{\sigma}) \qquad \bar{\sigma}^\wedge = (1, -\vec{\sigma})$$

The Dirac equation is:

$$\begin{pmatrix} -m & i\sigma\cdot\partial \\ i\bar{\sigma}\cdot\partial & -m \end{pmatrix}\begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

So the Weyl equations are:

$$i\bar{\sigma}\cdot\partial\psi_L = 0 \qquad i\sigma\cdot\partial\psi_R = 0$$

While we'll focus on the Dirac theory (appropriate for QED and QCD) for now, the electroweak theory will involve <u>chiral</u> (two-component, Weyl) spinors.